Matrix Calculus

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基本概念

标量 | 只有大小，没有方向，可用实数表示的量.

实值函数 | 函数$y=f(x), y\in\mathbb{R}, x\in\mathbb{C}$

梯度 | 根据自变量和因变量的不同可以分为：

✓ 自变量为实向量的标量函数关于向量的梯度.

$$f\in\mathbb{R}, \nabla_x f=[\frac{\partial f}{\partial x_1},\cdots,\frac{\partial f}{\partial x_n}]^T=\frac{\partial f}{\partial x}.$$

✓ 自变量为实向量的向量函数关于向量的梯度.

$$f\in\mathbb{R}^{1\times n}, \nabla_x f=\begin{bmatrix}\frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_1}\\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_2}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial f_1}{\partial x_n} & \frac{\partial f_2}{\partial x_n} & \cdots & \frac{\partial f_m}{\partial x_n}\end{bmatrix}=\frac{\partial f}{\partial x}.$$

✓ 自变量为矩阵的标量函数关于矩阵的梯度.

$$f\in\mathbb{R}, \nabla_X f=\begin{bmatrix}\frac{\partial f}{\partial x_{11}} & \frac{\partial f}{\partial x_{12}} & \cdots & \frac{\partial f}{\partial x_{1n}}\\ \frac{\partial f}{\partial x_{21}} & \frac{\partial f}{\partial x_{22}} & \cdots & \frac{\partial f}{\partial x_{2n}}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial f}{\partial x_{m1}} & \frac{\partial f}{\partial x_{m2}} & \cdots & \frac{\partial f}{\partial x_{mn}}\end{bmatrix}=\frac{\partial f}{\partial X}.$$

Jacobian矩阵 | 若函数$f(x):\mathbb{R}^n\to\mathbb{R}^m$，有

$$x=[x_1,x_2,\cdots,x_n]^T, f(x)=\begin{bmatrix}f_1(x_1,\cdots,x_n)\\ \vdots\\ f_m(x_1,...,x_n)\end{bmatrix}.$$

其Jacobian矩阵$J(x)$可以写为

$$f\in\mathbb{R}^{m\times 1}, J(x)=\begin{bmatrix}\frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n}\\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n}\end{bmatrix}=\nabla_x^T f.$$

✓ Jacobian矩阵表现了向量函数的最佳线性逼近.

Hessian矩阵 | 若$f:\mathbb{R}^n\to\mathbb{R}$为二次可导函数，$x=[x_1,\cdots,x_n]^T$，$f$的Hessian矩阵为

$$H(x)=\begin{bmatrix}\frac{\partial^2 f}{\partial x_1\partial x_1} & \frac{\partial^2 f}{\partial x_1\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1\partial x_n}\\ \frac{\partial^2 f}{\partial x_2\partial x_1} & \frac{\partial^2 f}{\partial x_2\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2\partial x_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial^2 f}{\partial x_n\partial x_1} & \frac{\partial^2 f}{\partial x_n\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n\partial x_n}\end{bmatrix}=\nabla_x^T f.$$

✓ Hessian矩阵使用函数的二阶信息，常用于Newton法解决大规模的优化问题.

实值函数有关向量的梯度

函数f关于行向量x^T的梯度为

$$\frac{\partial f}{\partial x^T}=[\frac{\partial f}{\partial x_1},\cdots,\frac{\partial f}{\partial x_n}]=\nabla_{x^T}f(x).$$

✓ 若f为常数，那么对应的梯度为$\mathbf{0}$.

✓ 如果A和y与x无关，那么

$$\frac{\partial x^T Ay}{\partial x}=\frac{\partial x^T}{\partial x}Ay=Ay.$$

加法法则 | 若$f(x),g(x)$均为x的实值函数

$$\frac{\partial[pf(x)+qg(x)]}{\partial x}=p\frac{\partial f(x)}{\partial x}+q\frac{\partial g(x)}{\partial x}.$$

乘法法则 | 若$f(x),g(x)$均为x的实值函数

$$\frac{\partial[f(x)g(x)]}{\partial x}=f(x)\frac{\partial g(x)}{\partial x}+g(x)\frac{\partial f(x)}{\partial x}.$$

除法法则 | 若$f(x)$为向量x的向量值函数

$$\frac{\partial[f(x)/g(x)]}{\partial x}=\frac{1}{\partial g^2(x)}[g(x)\frac{\partial f(x)}{\partial x}-f(x)\frac{\partial g(x)}{\partial x}].$$

链式法则 | 若$g(x)$为x的向量值函数

$$\frac{\partial g(f(x))}{\partial x}=\frac{\partial f^T(x)}{\partial x}\frac{\partial g(f)}{\partial f}.$$

常见类型的实值函数的梯度为

$$\nabla(x^Tx)=\frac{\partial x^Tx}{\partial x}=2x^T$$
$$\nabla(a^Tx)=\frac{\partial a^Tx}{\partial x}=a, \nabla(x^TA)=\frac{\partial x^TA}{\partial x}=A$$
$$\nabla(x^TAx)=\frac{\partial x^TAx}{\partial x}=(A+A^T)x$$

矩阵迹、行列式的梯度矩阵

矩阵迹的性质 | 二次型函数的迹与它本身相等.

$$f(x)=x^TAx=\mathrm{Tr}(x^TAx)=\mathrm{Tr}(xx^TA).$$

✓ 有关矩阵的迹，常见的梯度计算公式：

$$\nabla(\mathrm{Tr}(X))=\frac{\partial\,\mathrm{Tr}(X)}{\partial X}=I$$
$$\nabla(\mathrm{Tr}(X^{-1}))=\frac{\partial\,\mathrm{Tr}(X^{-1})}{\partial X}=-(X^{-1})^T$$
$$\nabla(\mathrm{Tr}(X^TX))=\frac{\partial\,\mathrm{Tr}(X^TX)}{\partial X}=(2X^T)^T=2X$$
$$\nabla(\mathrm{Tr}(XA))=\frac{\partial\,\mathrm{Tr}(XA)}{\partial X}=\frac{\partial\,\mathrm{Tr}(AX)}{\partial X}=A^T$$

✓ 有关矩阵的行列式，常见的梯度计算公式：

$$\nabla(\det(X))=\frac{\partial\det(X)}{\partial X}=\det(X)(X^{-1})^T$$
$$\nabla(\det(X^{-1}))=\frac{\partial\det(X^{-1})}{\partial X}=-(\det(X))^{-1}(X^{-1})^T$$
$$\nabla(\det(XX^T))=\frac{\partial\det(XX^T)}{\partial X}=2\det(XX^T)(XX^T)^{-1}X$$
$$\nabla(\det(\log X))=\frac{1}{\det(X)}\frac{\partial\det(X)}{\partial X}=2X^{-1}-\mathrm{diag}(X^{-1})$$

实值函数的梯度矩阵

✓ $X\in\mathbb{R}^{m\times n}, f(X)=c\ \nabla_X f(X)=\mathbf{0}_{m\times n}$

加法法则 | 若$f(X),g(X)$均为矩阵X的实值函数

$$\frac{\partial[pf(X)+qg(X)]}{\partial X}=p\frac{\partial f(X)}{\partial X}+q\frac{\partial g(X)}{\partial X}.$$

乘法法则 | 若$f(X),g(X)$均为矩阵X的实值函数

$$\frac{\partial[f(X)g(X)]}{\partial X}=f(X)\frac{\partial g(X)}{\partial X}+g(X)\frac{\partial f(X)}{\partial X}.$$

除法法则 | 若$f(X),g(X)$为X的函数，$g(X)\neq 0$

$$\frac{\partial[f(X)/g(X)]}{\partial X}=\frac{1}{g^2(X)}[g(X)\frac{\partial f(X)}{\partial X}-f(X)\frac{\partial g(X)}{\partial X}].$$

链式法则 | 若$g(X)$是自变量为矩阵X的实值函数，$f(y)$是自变量为标量y的实值函数

$$\frac{\partial f(g(X))}{\partial X}=\frac{\partial f(y)}{\partial y}\frac{\partial g(X)}{\partial X}.$$

常见类型的实值函数的梯度矩阵计算公式

$$\nabla_X(a^TXy)=\frac{\partial a^TXy}{\partial X}=ay^T$$
$$\nabla_X(a^TXX^Ty)=\frac{\partial a^TXX^Ty}{\partial X}=(ay^T+ya^T)X$$
$$\nabla_X(e^{a^TXy})=\frac{\partial e^{a^TXy}}{\partial X}=ay^Te^{a^TXy}$$

标量函数的微分

✓ 标量函数$f(x)$的导数$f'(x)$定义为

$$f'(x)=\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

即$f(x+\Delta x)=f(x)+\Delta x f'(x)+R, \lim_{\Delta x\to 0}\frac{R}{\Delta x}=0.$

上式称为**泰勒公式的一阶展开式**.

✓ $f(x)$在x点的**一阶微分**为 $df(x)=\Delta x f'(x)$.

矩阵的微分

矩阵微分 | 矩阵微分是实函数微分对矩阵函数的推广情况.

如果$x,\Delta x$为$n\times 1$的向量，$\exists A(x)\in\mathbb{R}^{m\times n}$，使得

$$f(x+\Delta x)=f(x)+A(x)\Delta x+R.$$

其中$\lim_{\Delta x\to 0}\frac{R}{\|\Delta x\|_2}=0$，那么函数$f(X)$在向量$x$处的一阶微分向量为

$$df(x)=A(x)\Delta x.$$

$A(x)$称为向量函数$f(x)$的一阶**导数矩阵**；如果向量函数$f(x)$在c处可微，$u\in\mathbb{R}^{n\times 1}$

$$df(c)=[D(f(x))]u, D(f(x))\in\mathbb{R}^{m\times n}.$$

D_{ij}表示$f(x)$第i个元素关于c的第j个元素的偏导；$D(f(x))$实质是$f(x)$在c处的Jacobian矩阵.

梯度矩阵 | $f(x)$在c处的梯度矩阵为$(D(f(x)))^T$.

求解$m\times 1$的向量函数$f(c)$的梯度矩阵$\nabla f(X)$：

✓ 求解向量函数微分$df(c)$，获得Jacobian矩阵；

✓ 将Jacobian矩阵转置，得到梯度矩阵$\nabla f(c)$.

常数矩阵微分 | $dC=0$.

常数与矩阵的乘积 | $dcX=cdX$.

矩阵加和的微分 | $d(X+Y)=dX+dY$.

矩阵乘积的微分 | $d(XY)=(dX)Y+X(dY)$.

常用的矩阵微分计算公式

$$d(X^T)=(dX)^T$$
$$d(\mathrm{Tr}(X))=\mathrm{Tr}(dX)$$
$$d(X^{-1})=-X^{-1}(dX)X^{-1}$$
$$d(\ln(X))=X^{-1}dX$$
$$d(\det(X))=\det(X)\,\mathrm{Tr}(X^{-1}dX)$$

二阶微分矩阵 | 矩阵函数$f(X)$的二阶微分为

$$d^2f(X)=Tr(U(dX)^TV(dX))$$

$$d^2f(X)=Tr(U(dX)V(dX))$$

与之对应的Hessian矩阵可写为

$$H(f(X))=\frac{1}{2}(U^T\otimes V+U\otimes V^T)$$

$$H(f(X))=\frac{1}{2}K_{nm}(U^T\otimes V+V^T\otimes U)$$

其中，K_{nm}为交换矩阵.

Gao Yang